

Navigating AI in professional service firms

An analysis of Deloitte

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Introduction

Professional service firms stay competitive because clients usually lack expertise to assess advice quality internally (Darbi & Karni 1973). Thus, clients rely on firm reputation when choosing audit opinions, tax filings, or strategic advice (Maister 1993). Since outcome quality is hard to measure (Empson et al. 2015), behavior-based contracts like Deloitte's billable-hours model dominate (Eisenhardt 1989), rewarding agents' time as a performance proxy. Here, principals reward agents' time and activity as proxies for performance. On this basis, Deloitte has grown to 470,000+ employees and ~\$70B revenue (2025).

Machine Learning is quickly eroding that foundation. Brynjolfsson and McAfee (2017) identified ML as the most important general-purpose technology (GPT) in our era, comparable to the transformation brought by electricity or the steam engine. This is particularly significant for professional services, as ML addresses Polanyi's Paradox (i.e., 'we know more than we can tell'). ML algorithms can bypass this limitation by learning underlying patterns directly from data rather than relying on explicit instruction. Audit transaction sampling, tax compliance classification, and document reviewing are all examples of tasks involving mapping structured inputs into labeled outputs, which ML excels at (Becker & Olaoye, 2024; Frontiers in AI, 2025; Faulconbridge et al. 2026). These tasks have justified Deloitte's broad base of graduate professionals (Saci & Aliouat, 2026). ML can now learn to perform these tasks automatically.

Disruption hits the historical model on two fronts. First, *capability substitution*: structured, data-heavy base tasks are automated, reducing need for large graduate cohorts. Second, *information asymmetry erosion*: as AI outputs become auditable, clients gain transparency and can assess outcomes themselves. This shifts market power to clients and raises new questions about accountability and digital trust.

This paper explores how Deloitte should respond to, and what results from, the dual threat of capability substitution and information asymmetry erosion. The analysis identifies key economic forces, cognitive constraints, and market vulnerabilities, then evaluates contract and capability needs for management.

Part 1

The structural economics: two forces reshaping expertise markets

Yerramilli-Rao et al. (2025) identify two forces changing competition in expertise: an expanding body of knowledge, making it harder to lead everywhere (p. 75), and falling expertise access costs, lowering market entry barriers (p. 75). For Deloitte, this means staying at the expertise frontier while facing new entrants that can match capabilities without years of infrastructure investment.

To address these issues (Yerramilli-Rao et al. 2025, p. 81) posits three questions each firm in this position should answer:

1. Which aspects of the problem we now solve for customers will customers use AI to solve themselves?
2. Which types of expertise that we currently possess will need to evolve most if we are to remain ahead of AI's capabilities?
3. Which assets can we build or augment to enhance our ability to stay competitive as AI advances?

Applied to Deloitte: question one covers due diligence, tax compliance, and regulatory screening; handled by graduates. Question two requires judgment about AI's future: regulatory navigation, stakeholder management, and organizational change. Question three is about assets new entrants can't easily replicate: regulatory ties, brand, proprietary systems, historical data. While these questions appear answerable through analysis, Deloitte must also implement the change, overcoming barriers to do so. To better explain the gap between strategic analysis and organizational action, we use another theoretical lens.

The cognitive constraint

Strategy originates in both the world of cognition in the form of mental representations, values, heuristics and in the world of action; routines and activities (Gavetti & Rivkin 2007, p. 431). The two are closely linked: representations shape heuristics, which in turn generate the routines and activities that create outcomes (Gavetti & Rivkin 2007, p. 432).

Mental *representations* are management's conception of the firm's identity and competition, it's the "mental models or cognitive frameworks underlying an actor's thought process" (Gavetti & Rivkin 2007, p. 428). At Lycos, managers saw their firm as tech, while Yahoo viewed itself as media (Gavetti & Rivkin 2007, p. 428). This difference shaped hiring, focus, and growth logic. *Focal heuristics* are search rules derived from a representation. Lycos's tech focus led to a heuristic to "do whatever it takes to demonstrate effective search" (Gavetti

& Rivkin 2007, p. 428). This local rationality blinded them to the need for media reach and scale for survival.

Deloitte's representation is similar: value comes from billing human expertise arranged in a pyramid with a large graduate base at the bottom, partners on top (Nordenflycht, S. 2010, p. 168). This view creates focal heuristics: grow graduate cohorts, price time not outcomes, and see headcount growth as capability (Nordenflycht, S. 2010, p. 161). These have shaped compensation, contracts, and talent acquisition for decades. Gavetti and Rivkin call the slow realization that a representation is inadequate the *awakening* construct. Awakening is slow because the channels for receiving signals are shaped by the existing representation.

At Deloitte, there are signs of awakening: \$3B committed to AI and cloud (Jain, 2025), agentic platforms in development (Deloitte, 2025), and reduced graduate intake (Koopman, 2025). But investment and commitments alone doesn't mean a changed representation; it can fit the old model. The real test is if focal heuristics shift; like partner compensation decoupling from headcount, or contracts moving from time-based billing. Yerramilli-Rao et al. (2025) and Gavetti & Rivkin (2007) reveal the underlying mechanisms of Deloitte's challenge. Next, we examine the threats.

Part 2

The external threat: Deloitte as a newly vulnerable incumbent

Clemons, Gu and Lang (2002) developed their theory of newly vulnerable markets for pure information products: fully digitizable goods delivered electronically, once defended by physical assets. While professional services aren't pure information products, much of Deloitte's portfolio now functions similarly due to AI (Becker & Olaoye, 2024; Frontiers in AI, 2025; Faulconbridge et al. 2026). This analysis targets products once manual, that can now be automated by AI.

Newly vulnerable markets must be easy to enter, attractive to attack, and difficult to defend (Clemons et al. 2002, p. 21).

Markets become newly easy to enter through regulatory, technological, or consumer change (Clemons et al. 2002, p. 22). AI tools now let boutique firms create structured audit and document reviews that once needed large teams and costly infrastructure. Commercially available technology removes the graduate expertise pipeline barrier to entry (Intuition Labs Editorial Team, 2025).

Attractive to attack means a strong customer profitability gradient (CPG), defined as "the presence of extreme differences in profitability between the best and the worst customers in a market" (Clemons et al. 2002, p. 22). New entrants can target high-profit clients and ignore low-profit ones.

Clemons et al. also note, "cross-subsidies can come from one product line being used to subsidize another, rather than one customer group" (Clemons et al. 2002, p. 22). Deloitte faces two exposures: profitable mandates offset lower-margin accounts (Iacone, 2025), and high-volume standardized work subsidizes less scalable advisory (Iacone, 2025). AI lets boutiques deliver high-margin standard work cheaply, avoiding unprofitable clients and services Deloitte must keep to retain its full service position.

Difficult to defend means barriers block incumbents from copying attackers. Clemons et al. point to "fixed commitments to existing customers to maintain current prices, and investments in inappropriate systems or physical infrastructure" (Clemons et al. 2002, p.23). Deloitte's main barrier is organizational: partner compensation tied to headcount, multiyear graduate commitments, and time-based client contracts. Decades-old structures were built for the old model. This is the Gavetti and Rivkin (2007) representation-rigidity problem: the very heuristic that built Deloitte's success now hinders adapting to low-cost attackers. Layering

Gavetti and Rivkin's cognitive angle onto Clemons et al. explains why deep-rooted problems may not be solved by investment alone.

The three conditions make Deloitte vulnerable, but why hasn't this vulnerability resulted in full scale disruption? First, accumulated switching costs (McIntyre & Chintakananda, 2014) mean clients face real transition costs. Internal processes, staff training, and existing relationships are slow to change. Second, Clemons, Reddi and Row (1993) show that relationship-specific capital requires time to recoup, which explains why even buyers who recognize the opportunity of switching may delay acting on it. Furthermore, for regulated services, such as auditing, increasing litigation risks and insurance costs dissuade smaller firms from entry (OECD, 2010). These defenses buy time but don't remove the threat from boutique entrants.

A key question is what assets Deloitte can use to counter attackers, connecting to Yerramilli-Rao et al. 2025's third question: "Which assets can we build or augment to enhance our ability to stay competitive as AI advances?" It's natural to assume Deloitte's historical data could give it an edge in building better models than new competitors.

Why the data moat fails

Agrawal, Gans and Goldfarb (2020) explain why this strategy is problematic. Competitive advantage in ML comes from two sources: access to sufficient training data and the speed of the feedback loop that maps outcomes back to the inputs that generated them, which is the mechanism that enables the algorithm to improve (Agrawal et al. 2020, p. 131). On the first dimension, client confidentiality prevents Deloitte from pooling outcome data across engagements, fragmenting what would otherwise be a substantial historical dataset. This data limitation impedes Deloitte's ability to train robust, generalizable models and diminishes the potential to derive meaningful insights from past projects. On the second, feedback in consulting is slow and often unverifiable: an audit opinion is only tested when a misstatement eventually surfaces, and the effect of strategic advice may take years to materialise. These limitations result in less effective machine learning model development and slower organizational learning, as models cannot be continually improved with timely feedback. Thus, Deloitte's perceived data advantage may be overstated, and its vulnerability in the face of competitors is driven more by underlying structural data constraints than by management's choices or AI implementation strategies.

Part 3

The contract consequence: from information asymmetry to a differentiated architecture

The billable-hours model is a behavior-based contract, justified by Eisenhardt's (1989) Proposition 9: "Outcome measurability is negatively related to behavior-based contracts and positively related to outcome-based contracts" (p. 62). Professional services are historically hard to observe and evaluate, so using billable hours as a proxy for performance makes sense.

AI disrupts this environment, but unevenly across Deloitte's portfolio. For structured tasks, AI produces auditable, specifiable outputs like compliance checklists or reviewed contracts, making outcomes measurable. Proposition 9 predicts this shifts preference to outcome-based pricing. Competitive pressure reinforces this, as boutique firms can offer similar outputs, weakening the case for billable hours.

For complex advisory, informational conditions don't change as much. Two factors support behavior-based contracting: outcome uncertainty stays high as results depend on external factors (Proposition 4); outcome measurability stays low as judgement quality is hard to assess (Proposition 9). Proposition 3 adds that AI-generated audit trails make advisory work more observable, letting Deloitte prove correct processes even when outcomes are unknowable. As clients can now verify processes, behavior-based contracts become more appealing. Here, AI reinforces rather than weakens this model.

A uniform billable-hours model is no longer relevant for Deloitte. For structured, specifiable services, Eisenhardt's propositions indicate clients will move to outcome-based pricing. Clemons, Reddi, and Row (1993) argue that IT reduces coordination costs and, crucially, transaction risks by increasing the standardization and interconnection of digital assets. For automated workflows, this reduces 'relationship-specificity,' lowering switching costs and allowing these services to be governed by market-like competition. Conversely, complex advisory requires high levels of explicit coordination and 'non-contractible' human assets such as trust and deep regulatory expertise. In these cases, IT's ability to lower coordination costs incentivizes a 'Move to the Middle', where clients concentrate their spend with a few long-term partners to protect these specialized, non-replicable assets. This allows Deloitte to differentiate its strategy: competing on scale and price for automated tasks while defending its advisory business through deep, partnership-based incumbency.

This split has important social effects. When AI produces audit work the lead partner signs off on results created without the usual human involvement. The EU AI Act (Regulation (EU) 2024/1689) does not call all auditing 'high-risk,' but AI tools used for important financial risk checks may require strong human oversight and transparency. As a result, outcome-based contracts should go beyond just listing deliverables and also explain how 'meaningful' human involvement happens. If not, firms may end up shifting the technical and knowledge responsibility to clients who cannot judge the AI's logic. This is the information asymmetry problem inverted: clients can now easily check results, but because the process is a 'black box,' neither auditors nor clients can fully prove how the audit was done.

Part 4

The internal capability: building AI as an organizing system

A differentiated contract model alone isn't enough for Deloitte to address a vulnerable market. To lead in expertise, they must manage AI as a capability, not just a plug-and-play tool (Stelmaszak, Joshi, & Constantiou, 2025). AI is "a connective, codependent, and emergent capability of a system of relations – among human and algorithmic actors – to organize by analysing, learning, and acting in pursuit of organizational goals" (Stelmaszak et al. 2025, p. 337). Deloitte risks treating AI as a product, expecting value from deployment alone, but the capability view requires managing three key properties.

Connectivity: "AI only arises as an organizing capability through connectivity, when the human and algorithmic actors enact relations to achieve organizational goals" (Stelmaszak et al. 2025, p. 347). The key is enactment, not mere presence: auditors who challenge AI-flagged anomalies enact the relationship, while passive approval doesn't count. This is vital for behavior-based contracts, where customers pay for professional judgment, not just clicking approve. Simply deploying a tool without specifying roles and purposes fails to create true connectivity. This is also vital for the responsible use of AI, humans must be in the loop to ensure AI isn't biased or causes undue harm (Regulation (EU) 2024/1689, 2024).

Codependence: "AI is codependent on the participating actors, humans as well as algorithms", but codependence "does not suggest equidependence" (Stelmaszak et al. 2025, p. 348). AI's role is greater for structured tasks, while humans dominate in complex ones, but both are essential. Deloitte must manage these relationships, not just acquire platforms.

Emergence: "AI emerges over time in practice as relations are enacted in a system" (Stelmaszak et al. 2025, p. 349). Capability evolves as humans and algorithms interact and learn. Senior partners must recognize AI isn't static; organizational capabilities must adapt as the system changes.

This workforce dimension to AI integration adds complexity to the investment decision in AI. Neeley and Leonardi (2022) establish that the development of a digital mindset, defined as "a set of attitudes and behaviors that enable people and organizations to see how data, algorithms, and AI open up new possibilities" (p. 51), depends on two dimensions. The first, "buy in": the degree to which employees believe the transformation produces benefits to them and the organization (p.52). The second, "capacity to learn": the degree to which they are confident they can acquire the needed technological competence (p.52). The dimensions create four quadrants: frustrated (high buy-in/low confidence), inspired (high buy-in/high confidence), indifferent (high confidence/low buy-in), and oppressed (low on both). When applying the matrix to Deloitte, there is a real risk at both the junior and the more senior

level. Juniors hired and trained for the structured work now being automated face a frustrated-quadrant risk: they can see the logic behind the transformation but the competencies that landed them their position is the very competencies being devalued, potentially undermining confidence that they can acquire the replacement skill set. Senior partners, on the other hand, face an indifferent-quadrant risk. Their confidence is typically high as they have transcended prior technological shifts, but buy-in may be compromised as the existing pyramid was the environment in which they achieved their status and role. This is the representation-rigidity problem identified by Gavetti and Rivkin (2007) applied to the individual level. The focal heuristic that made a partner successful could now hinder their ability to reconfigure the organization effectively.

Neeley and Leonardi (2022) prescribe the medicine: a bold stroke commanding attention must be followed by a “long march” (p. 52) of sustained change management. Deloitte’s pivot is the bold stroke, the sustained effort with prolonged reskilling, revised partnership tracks, new compensation structures, is the long march. This long march is what Gavetti and Rivkin’s (2007) cognitive constraint predicts will be stiff and often come too late.

Part 5

Consequences and managerial recommendations

Using theories of market forces, cognitive limits, market vulnerability, contract logic, and the capability view, the path forward is more than just investing in AI; it requires full firm re-engineering.

Addressing capability redistribution

Capability redistribution collapses Deloitte's traditional pyramid structure. As shown by Yerramilli-Rao et al. and Clemons et al., structured bottom-tier tasks are now most replaceable and exposed. Deloitte must unlink partner compensation from headcount growth. Per Gavetti & Rivkin's (2007) awakening construct, the firm needs to shift from a labor- to capital-intensive mental model.

To meet Stelmaszak et al.'s (2025) connectivity standard, junior roles must shift from routine audit/tax tasks to AI contextualizers, trained to spot AI hallucinations and handle edge cases. In this way, junior employees act as moderators and ethical stewards of AI outputs, helping to prevent automation bias and ensuring that digital decisions align with professional standards and societal values (Regulation (EU) 2024/1689, 2024). This matters beyond Deloitte: professional services firms function as accountability anchors in markets. Ceding that function to AI without governance structures would undermine the social license on which the entire profession rests. This transition not only addresses concerns about job loss from AI replacement but repositions junior talent as key contributors to digital trust and the responsible use of AI. With proper incentives, the frustrated risk for juniors (Part 4) can be reduced by showing clear paths where AI can't replace them.

Solving for information asymmetry erosion

As information asymmetry erodes, clients can now use AI to audit Deloitte, no longer relying solely on reputation. For structured services, Deloitte should move to outcome-based pay (Eisenhardt, 1989), with fixed-fee contracts tied to auditable deliverables that clients can independently verify. For complex advisory, retain behavior-based billing and add AI audit trails to prove diligence. By competitively pricing standardized tasks and bundling them with trusted advisory, Deloitte can offer simpler services as add-ons while upholding new social norms around digital accountability. Deloitte should further position this transparency as a market differentiator: firms willing to disclose AI involvement and demonstrate human oversight can build the digital trust that eroding information asymmetry now requires of all professional service providers.

Leveraging the regulatory moat

To address Yerramilli-Rao et al.'s (2025) third question on staying competitive as AI advances, Deloitte should leverage its credibility and experience. The EU AI Act creates compliance needs (Regulation (EU) 2024/1689, 2024), shielding against vulnerabilities

(Clemons, Gu & Lang, 2002). Deloitte's regulatory expertise and credibility are hard to copy, turning AI from a substitute into a subject of its work.

The "Long March"

The analysis reveals that Deloitte's challenge is not primarily technological but structural. Yerramilli-Rao et al.'s (2025) three questions map to three interacting constraints: the what to automate question exposes Deloitte's cross-subsidies; the what expertise to evolve question is blocked by Gavetti and Rivkin's (2007) cognitive rigidity; and the which assets to build question is undermined by the data moat failure Agrawal et al. (2020) predict. These are not independent problems, they compound. Cognitive rigidity delays the contract transition Eisenhardt (1989) prescribes, which in turn delays the revenue shift needed to fund the AI capability Stelmaszak et al. (2025) require. The "Long March" is difficult precisely because each delay makes the next step harder. For Deloitte, the window is not permanently open: switching costs and barriers buy time, but as Clemons et al. (2002) show, newly vulnerable markets eventually tip.

External References:

Darbi, M. U., & Karni, E. (1973). "Free Competition and the Optimal Amount of Fraud." *Journal of Law and Economics*.

Maister, D. H. (1993). *Managing the Professional Service Firm*.

Empson, L., Muzio, D., Broschak, J., & Hinings, B. (2015). *The Oxford Handbook of Professional Service Firms*.

Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act). Official Journal of the European Union, L 2024/1689.

<https://eur-lex.europa.eu/eli/reg/2024/1689/oj/eng>

Becker, S., & Olaoye, O. (2024). *Enhancing audit quality and reducing costs: The impact of AI in banking and financial services*. PMC.

<https://pmc.ncbi.nlm.nih.gov/articles/PMC12876229/>

Frontiers in AI. (2025). Minimizing unnecessary tax audits using multi-objective hyperparameter tuning of XGBoost with focal loss. *Frontiers in Artificial Intelligence*, 8. <https://www.frontiersin.org/journals/artificial-intelligence/articles/10.3389/frai.2025.1669191/full>

Oxford Academic. (2026). Generative machine learning in professional work and professional service firms: A research agenda. *Journal of Professions and Organization*, 13(2). <https://academic.oup.com/jpo/article/13/2/joaf018/8519415>

Organisation for Economic Co-operation and Development. (2010). Competition and regulation in auditing and related professions.

Saci, F., & Aliouat, B. (2026). *How the Adoption of Artificial Intelligence Is Changing the Nature of Work in Accounting Firms? In Accounting in the 21st Century - Digital Transformation, Risk Governance, and Sustainability*. IntechOpen. <https://doi.org/10.5772/intechopen.1223934>

Von Nordenflycht, S. (2010). "What is a professional service firm? Toward a theory and taxonomy of knowledge-intensive firms." *Academy of Management Review*.

Jain, S. (2025, June 10). *Deloitte hits \$70.5bn global revenue; big bet on autonomous AI co-workers*. *The Finance Story*. <https://thefinancestory.com/deloitte-hits-70-5bn-global-revenue-big-bet-on-autonomous-ai-co-workers>

Deloitte. (2025, May 27). *Deloitte launches Global Agentic Network to power digital workforce solutions.*

<https://www.deloitte.com/global/en/about/press-room/deloitte-launches-global-agentic-network-to-power-digital-workforce-solutions.html>

Koopman, S. (2025, June 23). *Big Four slash graduate jobs as AI takes on entry level work.* City AM.

<https://www.cityam.com/big-four-slash-graduate-jobs-as-ai-takes-on-entry-level-work/>

Intuition Labs Editorial Team. (2025, May 22). *The impact of AI on graduate jobs in 2025.* Intuition Labs. <https://intuitionlabs.ai/articles/ai-impact-graduate-jobs-2025>

Iacone, A. (2024, May 1). *Big Four firms embrace AI to revamp corporate service offerings.* Bloomberg Tax.

<https://news.bloombergtax.com/financial-accounting/big-four-firms-embrace-ai-to-revamp-corporate-service-offerings>

Course Literature References:

Eisenhardt, K. M. (1989). Agency theory: An assessment and review. *The Academy of Management Review*, 14(1), 57–74. <https://doi.org/10.2307/258191>

Brynjolfsson, E., & McAfee, A. (2017). The business of artificial intelligence: What it can — and cannot — do for your organization. *Harvard Business Review*.

Yerramilli-Rao, B., Corwin, J., Li, Y., & Lakhani, K. R. (2025, January–February). Strategy in an era of abundant expertise: How to thrive when AI makes knowledge and know-how cheaper and easier to access. *Harvard Business Review*, 72–81.

Gavetti, G., & Rivkin, J. W. (2007). On the origin of strategy: Action and cognition over time. *Organization Science*, 18(3), 420–439. <https://doi.org/10.1287/orsc.1070.0282>

Clemons, E. K., Gu, B., & Lang, K. R. (2002). Newly vulnerable markets in an age of pure information products: An analysis of online music and online news. *Journal of Management Information Systems*, 19(3), 17–41.

McIntyre, D. P., & Chintakananda, A. (2014). Competing in network markets: Can the winner take all? *Business Horizons*, 57(1), 117–125. <https://doi.org/10.1016/j.bushor.2013.09.005>

Agrawal, A. , Gans, J. , & Goldfarb, A.. (2020, September–October) . How to win with machine learning: And how to catch up if you're lagging behind. *Harvard Business Review* , 126–133.

Neeley, T., & Leonardi, P. (2022, May–June). Developing a digital mindset: How to lead your organization into the age of data, algorithms, and AI. *Harvard Business Review*, 100(3), 50–55.

Clemons, E. K., Reddi, S. P., & Row, M. C. (1993). The impact of information technology on the organization of economic activity: The "move to the middle" hypothesis. *Journal of Management Information Systems*, 10(2), 9–35.